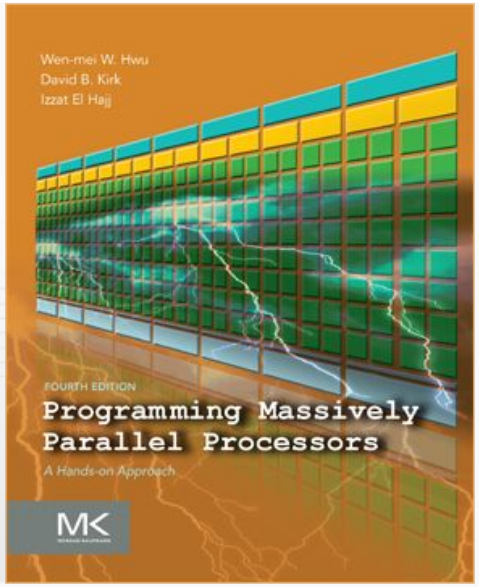
Programming Massively Parallel Processors

A Hands-on Approach

CHAPTER 11 › Prefix Sum (Scan)



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Scan

- A **scan** operation:
  - Takes:
    - An input array $[x_0, x_1, \ldots, x_{n-1}]$
    - An associative operator $\oplus$
      - e.g., sum, product, min, max
  - Returns:
    - An output array $[y_0, y_1, \ldots, y_{n-1}]$ where
      - **Inclusive scan**: $y_i = x_0 \oplus x_1 \oplus \ldots \oplus x_i$
      - **Exclusive scan**: $y_i = x_0 \oplus x_1 \oplus \ldots \oplus x_{i-1}$

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Scan Example

- Addition example:

3	6	7	4	8	2	1	9

↓

3	9	16	20	28	30	31	40

Inclusive Scan

3	6	7	4	8	2	1	9

↓

0	3	9	16	20	28	30	31

Exclusive Scan

- In general:

x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7

↓

x_0	$x_0..x_1$	$x_0..x_2$	$x_0..x_3$	$x_0..x_4$	$x_0..x_5$	$x_0..x_6$	$x_0..x_7$

Inclusive Scan

x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7

↓

ID	x_0	$x_0..x_1$	$x_0..x_2$	$x_0..x_3$	$x_0..x_4$	$x_0..x_5$	$x_0..x_6$

Exclusive Scan

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Sequential Scan

- Sequential scan for sum:

```
output[0] = input[0];
for(i = 1; i < N; ++i) {
    output[i] = output[i-1] + input[i];
}
```

Inclusive Scan

```
output[0] = 0.0f;
for(i = 1; i < N; ++i) {
    output[i] = output[i-1] + input[i-1];
}
```

Exclusive Scan

- In general:

```
output[0] = input[0];
for(i = 1; i < N; ++i) {
    output[i] = f(output[i-1], input[i]);
}
```

Inclusive Scan

```
output[0] = IDENTITY;
for(i = 1; i < N; ++i) {
    output[i] = f(output[i-1], input[i-1]);
}
```

Exclusive Scan

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Segmented Scan

- Parallel scan requires synchronization across parallel workers

- Approach: **segmented scan**
  - Every thread block scans a segment
  - Scan the segments' partial sums
  - Add each segment's scanned partial sum to the next segment

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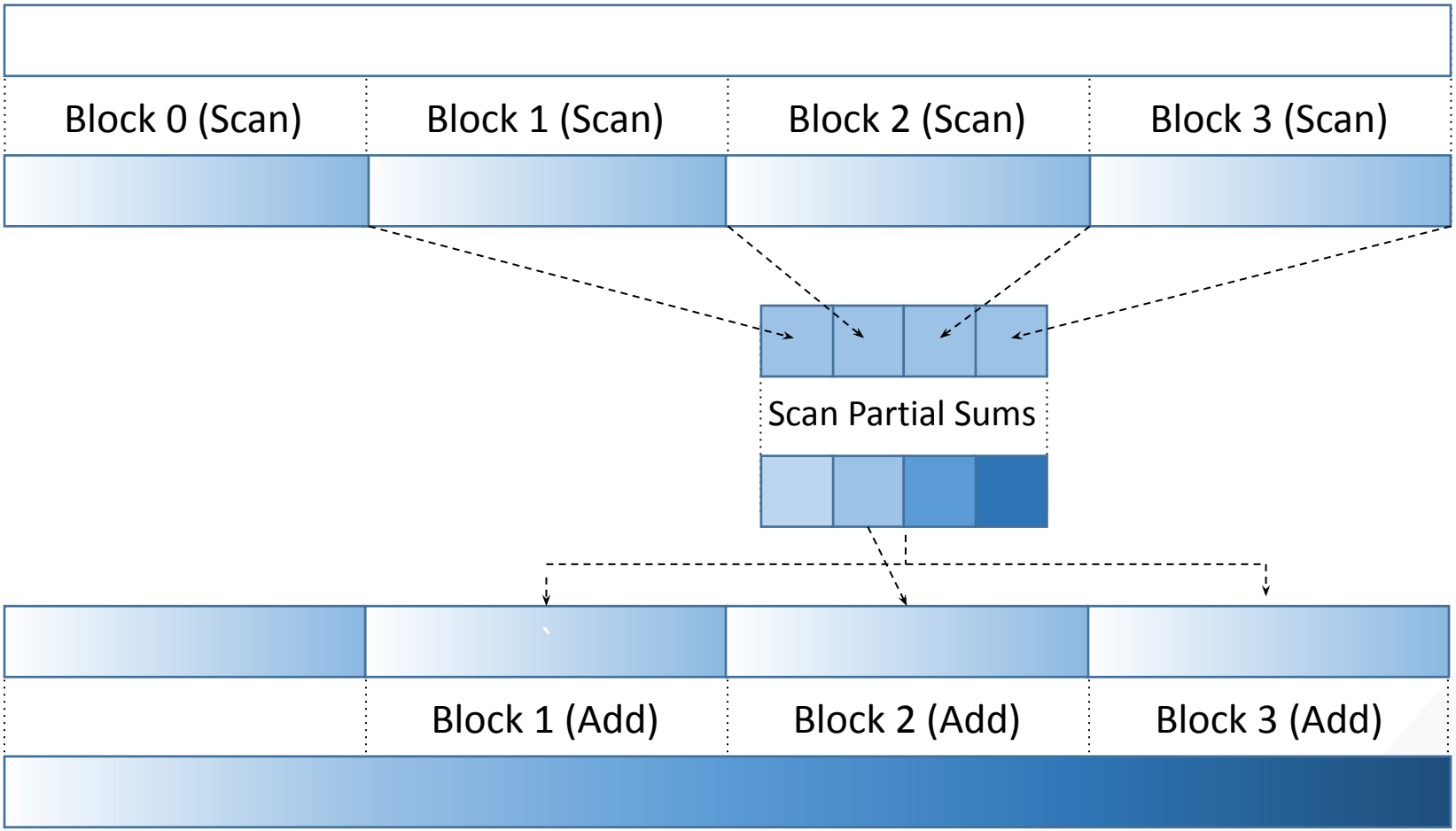
Segmented Scan Example



For now, we will focus on implementing a parallel scan in each block

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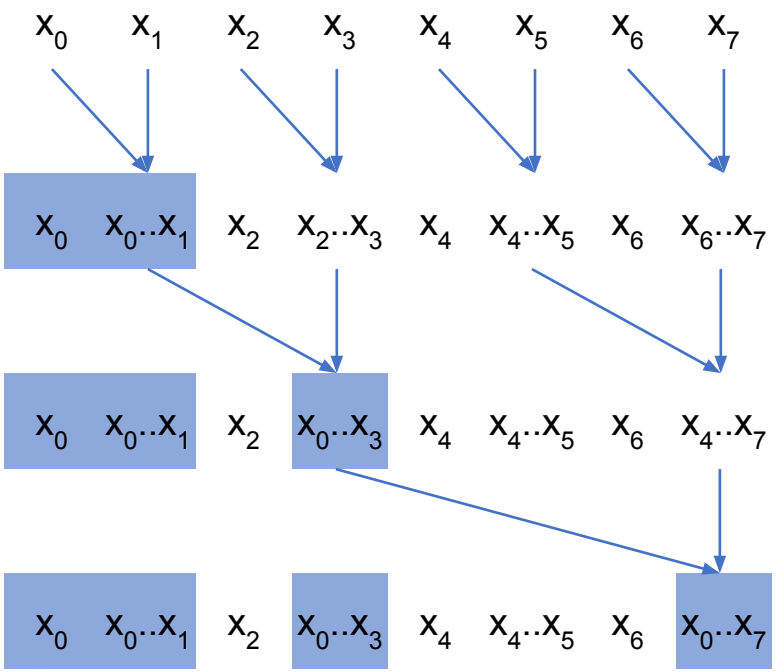
Parallel (Inclusive) Scan



A parallel reduction tree for the last element gives some others as a byproduct

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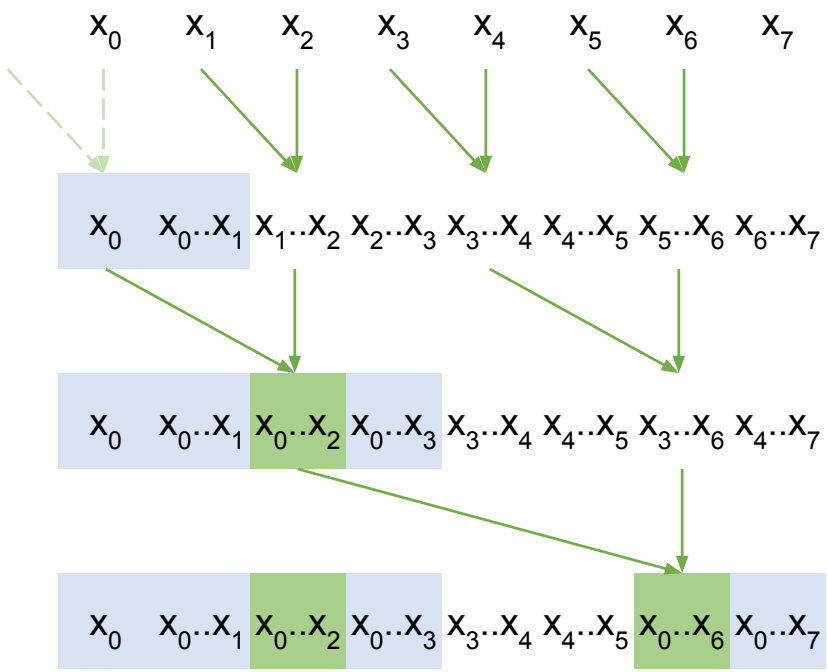
Parallel (Inclusive) Scan



Another reduction tree gives us more elements

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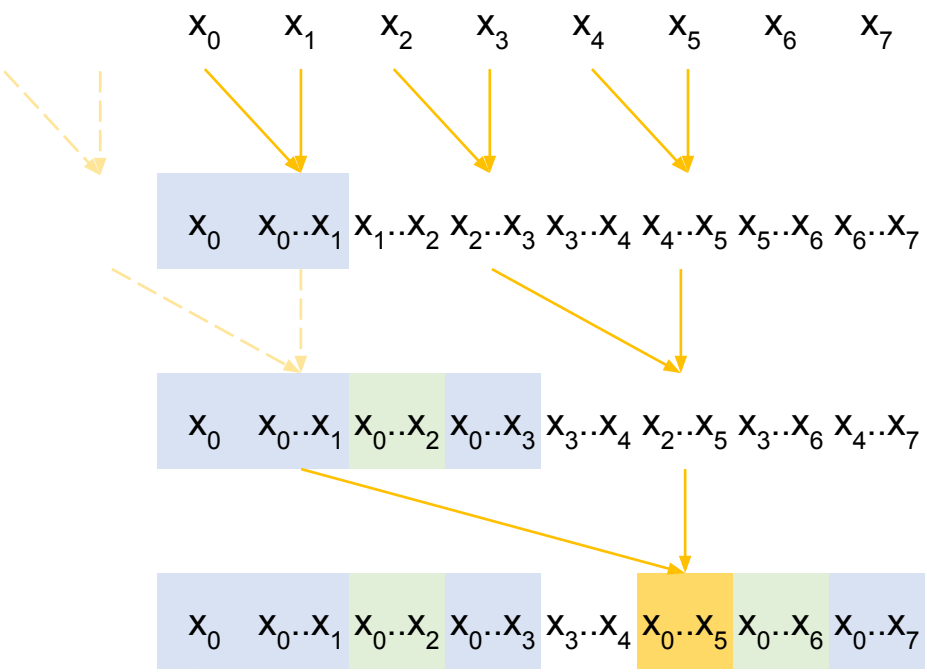
Parallel (Inclusive) Scan



Keep doing reduction trees until we get all answers

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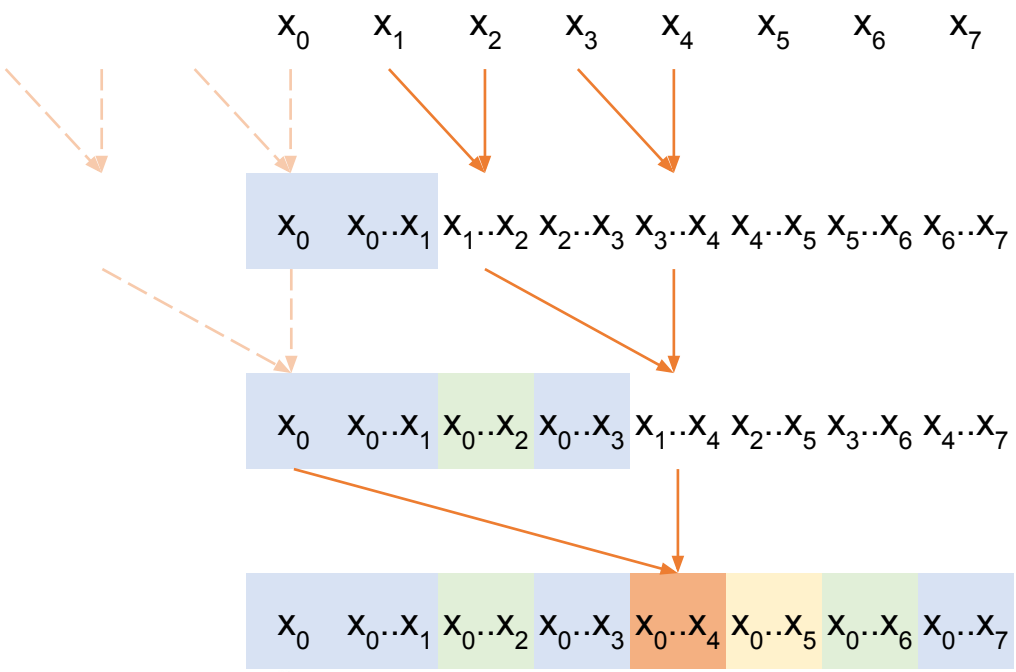
Parallel (Inclusive) Scan



Keep doing reduction trees until we get all answers

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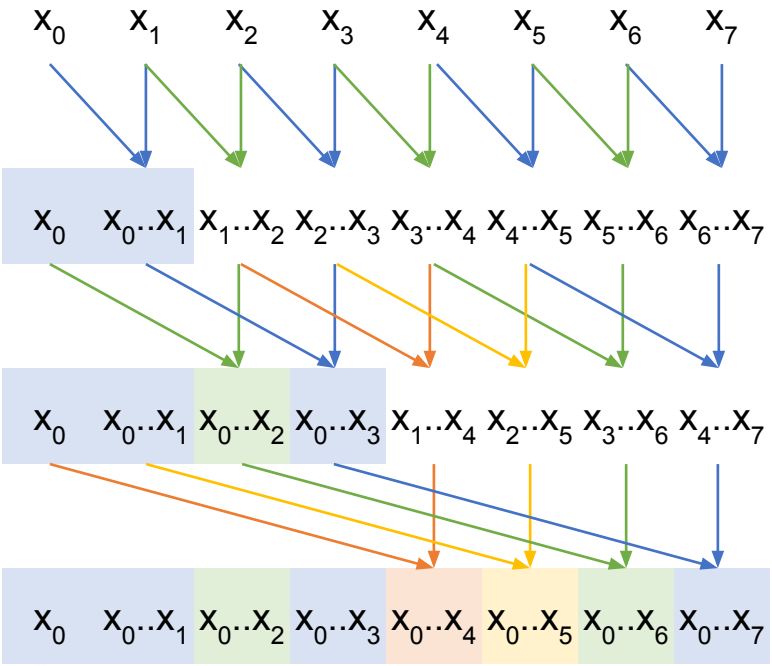
Parallel (Inclusive) Scan



Overlap the trees and do them simultaneously

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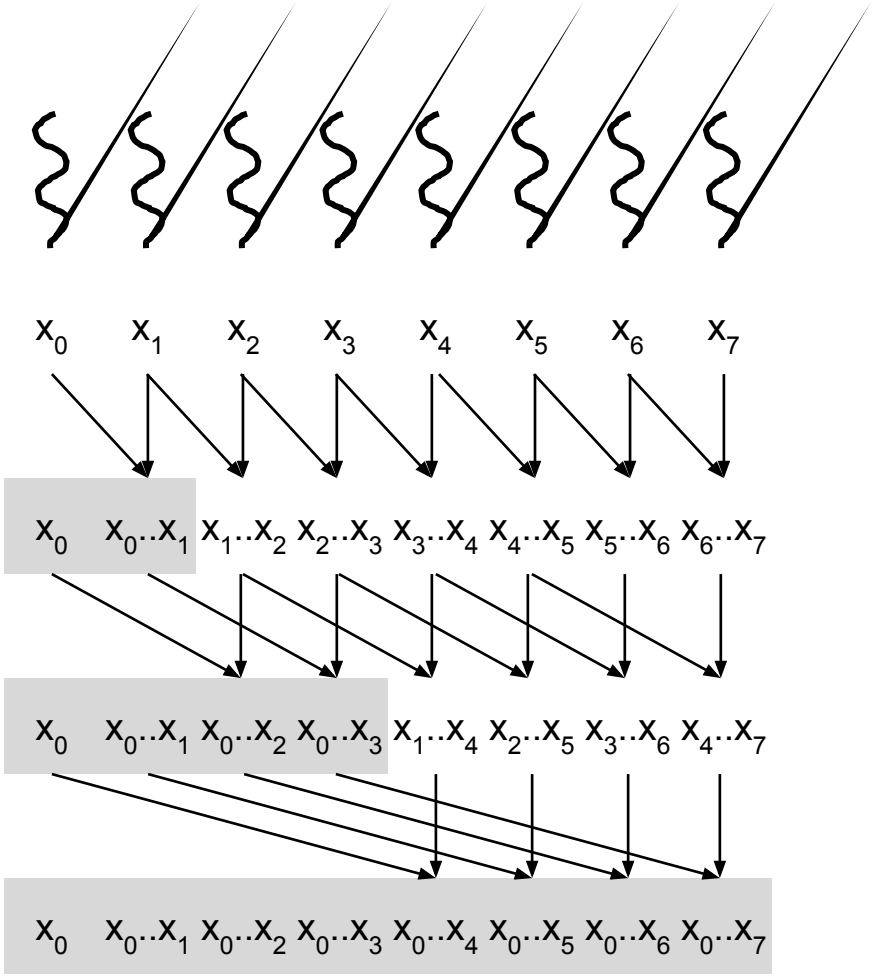
Kogge-Stone Parallel (Inclusive) Scan



One thread for each element

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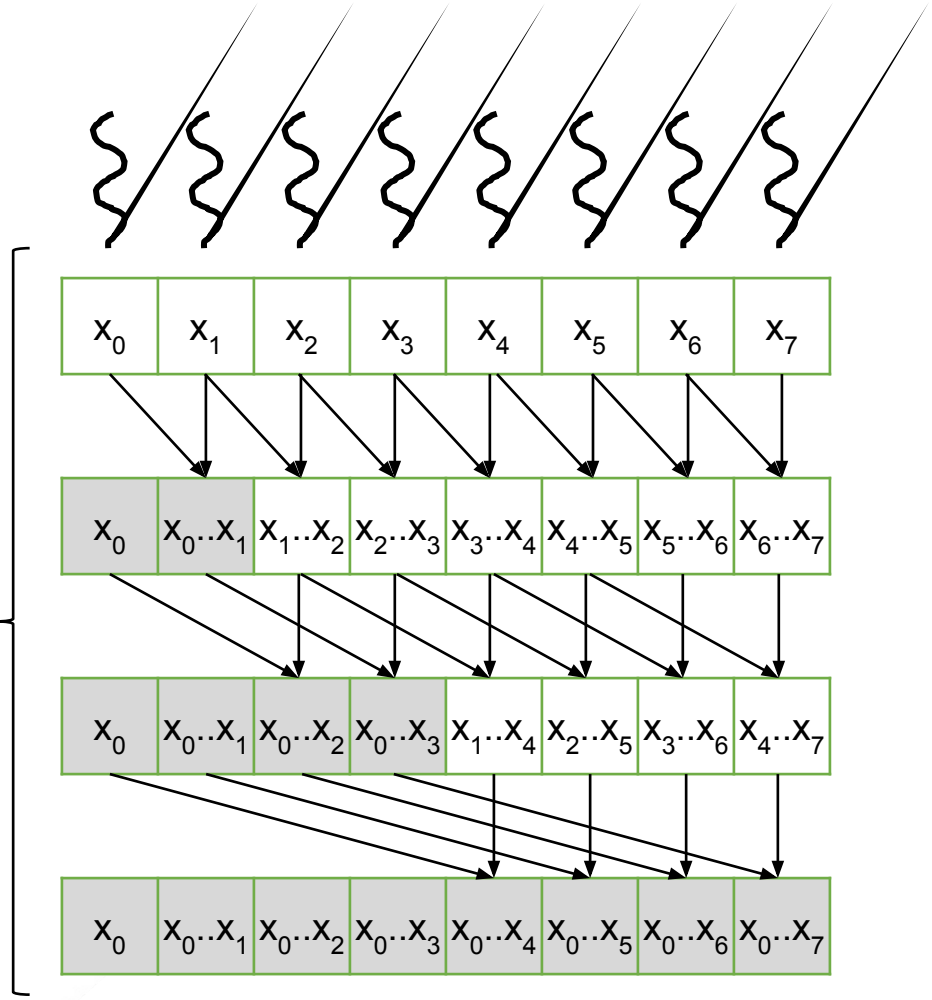
Using Shared Memory



One thread for each element

Optimization: load once to a **shared memory** buffer and perform successive reads and writes to the same array can be done in shared memory

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Kogge-Stone Parallel (Inclusive) Scan Code

```
unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;

__shared__ float buffer_s[BLOCK_DIM];
buffer_s[threadIdx.x] = input[i];
__syncthreads();

for(unsigned int stride = 1; stride <= BLOCK_DIM/2; stride *= 2) {
    if(threadIdx.x >= stride) {
        buffer_s[threadIdx.x] += buffer_s[threadIdx.x - stride];
    }
    __syncthreads();
}

if(threadIdx.x == BLOCK_DIM - 1) {
    partialSums[blockIdx.x] = buffer_s[threadIdx.x];
}

output[i] = buffer_s[threadIdx.x];
```

Incorrect!

Different threads are reading and writing the same data location without synchronizing

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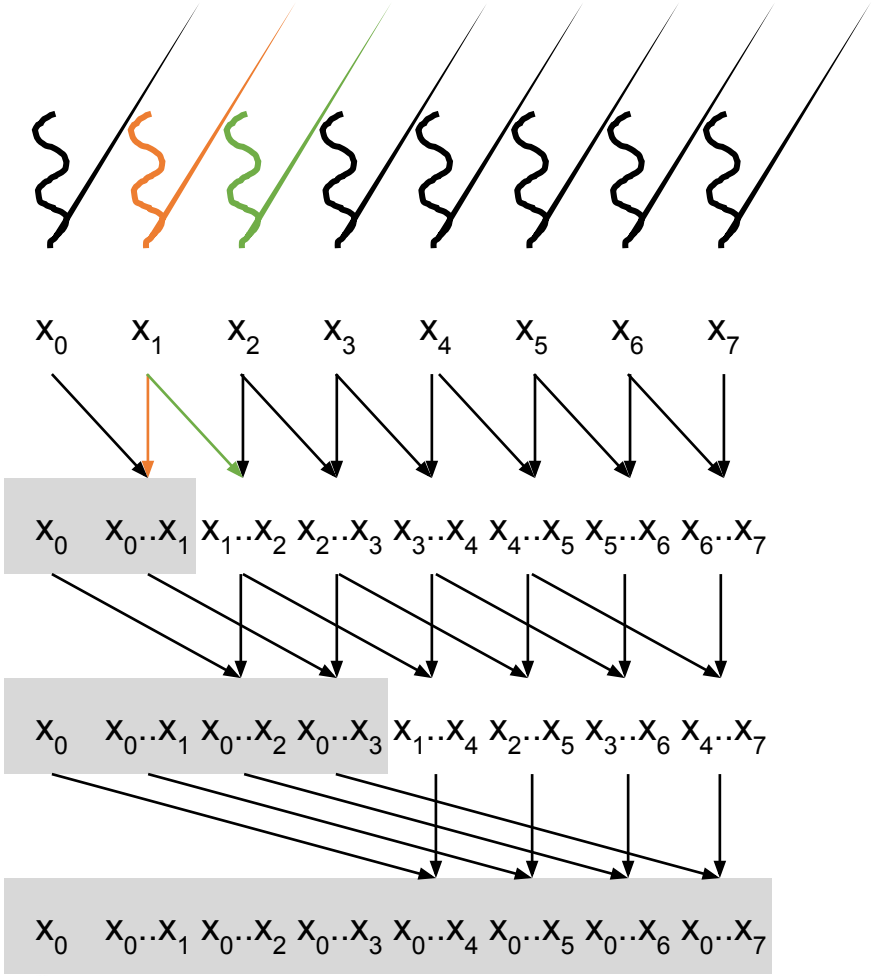
Kogge-Stone Parallel (Inclusive) Scan

Thread 1 may **update value at index 1** before **thread 2 reads it**

<u>Solution:</u> wait for everyone to read before updating



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Kogge-Stone Parallel (Inclusive) Scan Code

```
unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;

__shared__ float buffer_s[BLOCK_DIM];
buffer_s[threadIdx.x] = input[i];
__syncthreads();

for(unsigned int stride = 1; stride <= BLOCK_DIM/2; stride *= 2) {
    float v;
    if(threadIdx.x >= stride) {
        v = buffer_s[threadIdx.x - stride];
    }
    __syncthreads();
    if(threadIdx.x >= stride) {
        buffer_s[threadIdx.x] += v;
    }
    __syncthreads();
}

if(threadIdx.x == BLOCK_DIM - 1) {
    partialSums[blockIdx.x] = buffer_s[threadIdx.x];
}

output[i] = buffer_s[threadIdx.x];
```

Wait for everyone to read before writing

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True and False Dependences

```
unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;

__shared__ float buffer_s[BLOCK_DIM];
buffer_s[threadIdx.x] = input[i];
__syncthreads();

for(unsigned int stride = 1; stride <= BLOCK_DIM/2; stride *= 2) {
    float v;
    if(threadIdx.x >= stride) {
        v = buffer_s[threadIdx.x - stride];
    }
    __syncthreads();
    if(threadIdx.x >= stride) {
        buffer_s[threadIdx.x] += v;
    }
    __syncthreads();
}

if(threadIdx.x == BLOCK_DIM - 1) {
    partialSums[blockIdx.x] = buffer_s[threadIdx.x];
}

output[i] = buffer_s[threadIdx.x];
```

This synchronization enforces a **false dependence** (we only need to finish reading before others write because we are using the same buffer)

This synchronization enforces a **true dependence** (we must finish writing before others can read)

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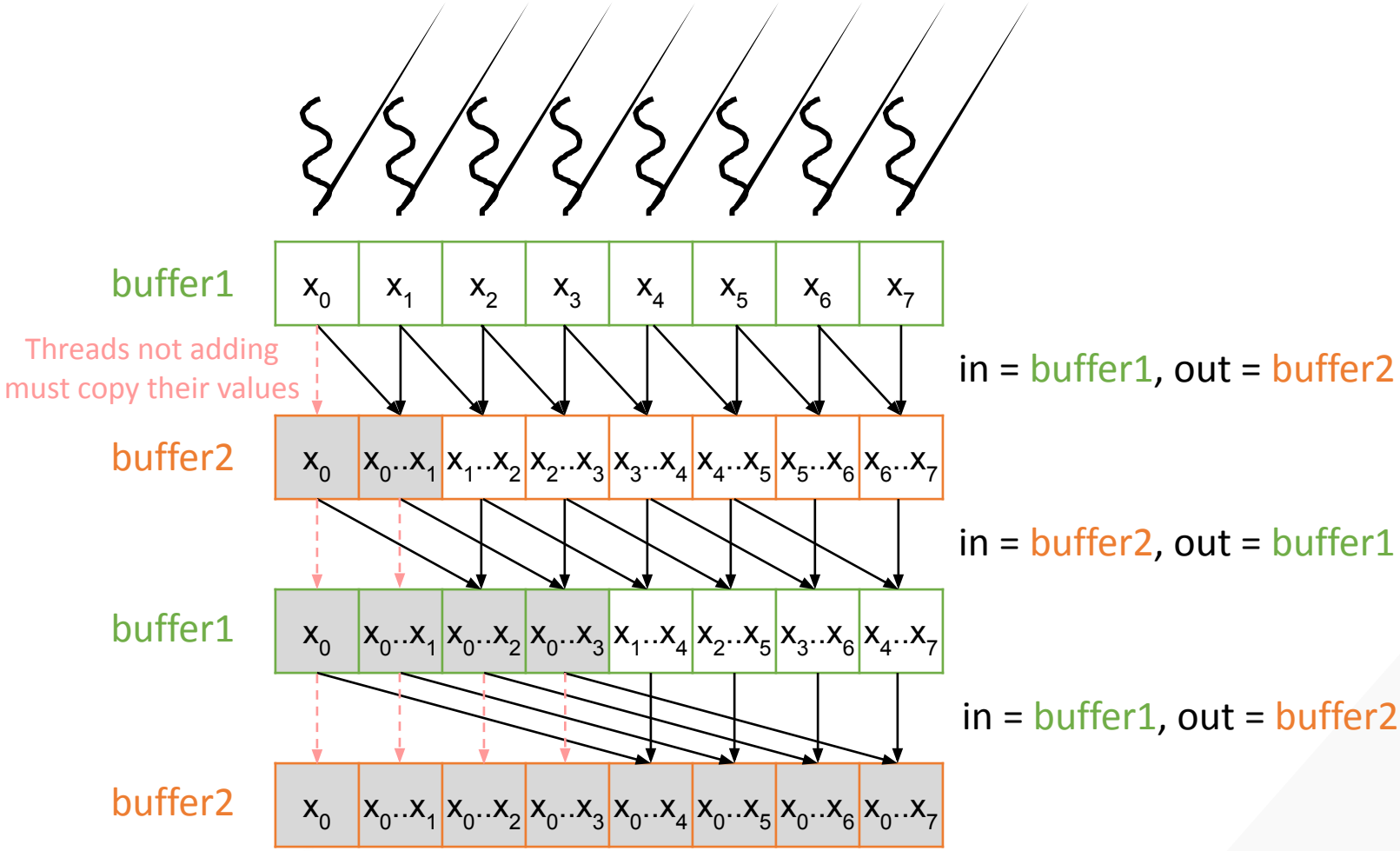
Double Buffering



Optimization: eliminate the synchronization that enforces a false dependence by using separate buffers for reading and writing, and alternate the buffers each iteration (called **double buffering**)

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Double Buffering Code

```
unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;

__shared__ float buffer1_s[BLOCK_DIM];
__shared__ float buffer2_s[BLOCK_DIM];
float* inBuffer_s = buffer1_s;
float* outBuffer_s = buffer2_s;
inBuffer_s[threadIdx.x] = input[i];
__syncthreads();

for(unsigned int stride = 1; stride <= BLOCK_DIM/2; stride *= 2) {
    if(threadIdx.x >= stride) {
        outBuffer_s[threadIdx.x] =
                inBuffer_s[threadIdx.x] + inBuffer_s[threadIdx.x - stride];
    } else {
        outBuffer_s[threadIdx.x] = inBuffer_s[threadIdx.x];
    }
    __syncthreads();
    float* tmp = inBuffer_s;
    inBuffer_s = outBuffer_s;
    outBuffer_s = tmp;
}

if(threadIdx.x == BLOCK_DIM - 1) {
    partialSums[blockIdx.x] = inBuffer_s[threadIdx.x];
}

output[i] = inBuffer_s[threadIdx.x];
```

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Work Efficiency

- A parallel algorithm is **work-efficient** if it performs the same amount of work as the corresponding sequential algorithm
- Scan work efficiency
  - Sequential scan performs *N* additions
  - Kogge-Stone parallel scan performs:
    - log(N) steps, N - 2^{step} operations per step
    - Total: (N-1) + (N-2) + (N-4) + … + (N-N/2)

      = N*log(N) - (N-1) = O(N*log(N)) operations
    - Algorithm is not work efficient
- If resources are limited, parallel algorithm will be slow because of low work efficiency

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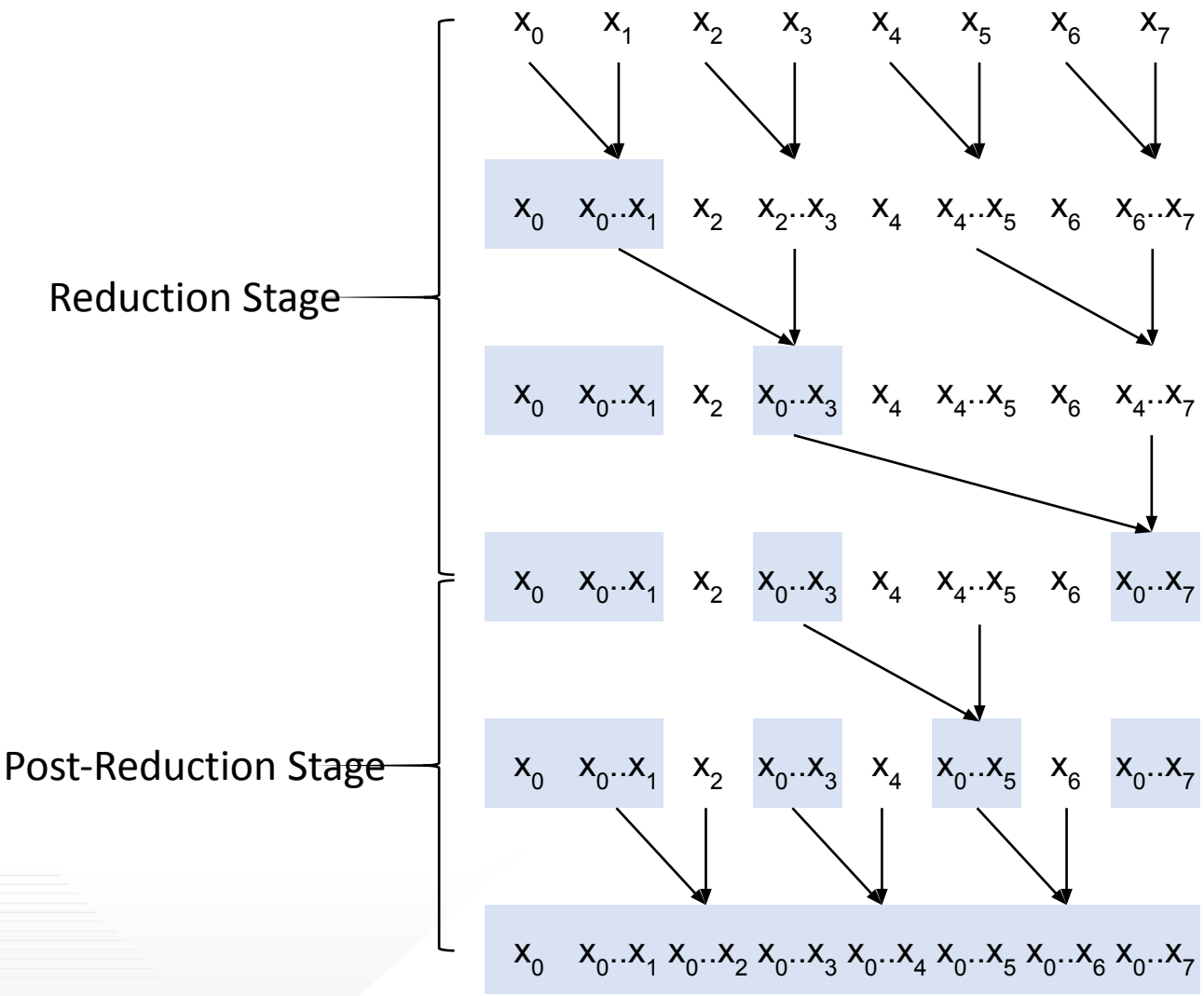
Brent-Kung Parallel (Inclusive) Scan



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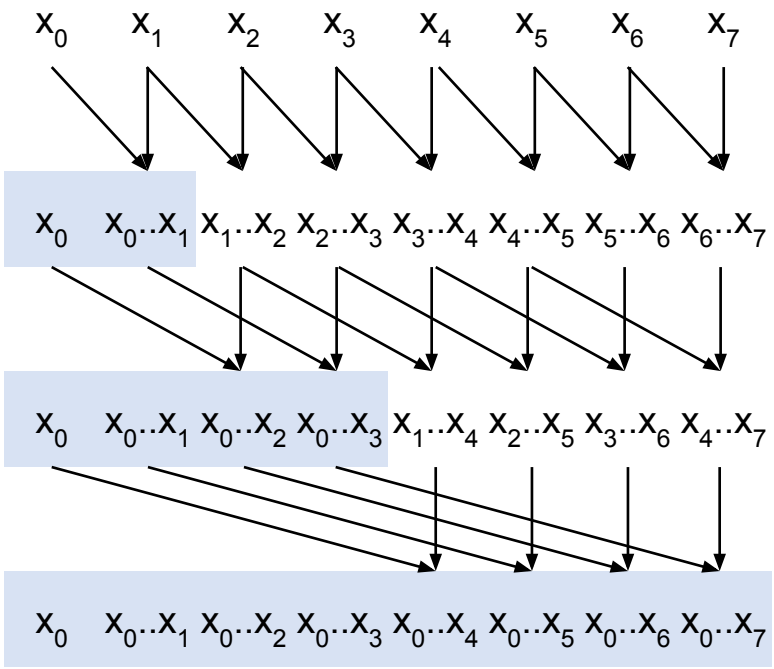
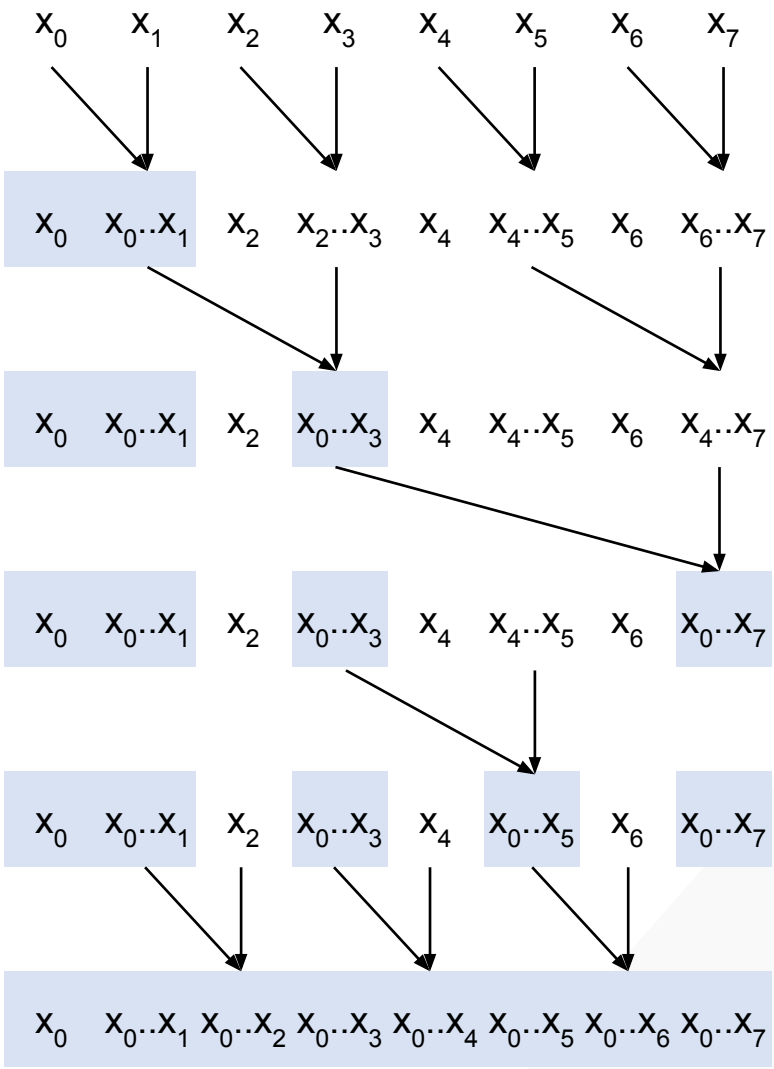
Kogge-Stone vs. Brent-Kung





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Work Efficiency

- Recall: Kogge-Stone
  - **log(N) steps**
  - **O(N*log(N)) operations**
- Brent-Kung
  - Reduction stage:
    - log(N) steps
    - N/2 + N/4 + … + 4 + 2 + 1 = N-1 operations
  - Post-Reduction stage:
    - log(N)-1 steps
    - (2-1) + (4-1) + … + (N/2-1) = (N-2) - (log(N)-1)
  - Total:
    - **2*log(N)-1 steps**
    - (N-1) + (N-2) - (log(N)-1) = 2*N – log(N) – 2 = **O(N) operations**
- Brent-Kung takes **more steps** but is **more work-efficient**
- So which one is faster?

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References

- Wen-mei W. Hwu, David B. Kirk, and Izzat El Hajj. *Programming Massively Parallel Processors: A Hands-on Approach*. Morgan Kaufmann, 2022.

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